Unit: 6 Unit 7: Factoring
Topic: Factoring by Grouping

Name: ________________
Date: ___________

Quiz (Carolina Reaper)

Q1. Factor the expression: $x^2 + 5x + 4x + 20$
- (A) $(x+4)(x+5)$
- (B) $(x+2)(x+10)$
- (C) $(x+1)(x+20)$
- (D) $(x+3)(x+6)$
- (E) $(x+5)(x+4)$

Q2. Factor the expression: $2x^2 + 5x + 3x + 7$
- (A) $(2x+3)(x+7)$
- (B) $(2x+7)(x+3)$
- (C) $(x+2)(2x+7)$
- (D) $(x+7)(2x+3)$
- (E) $(x+3)(2x+5)$

Q3. Factor the expression: $x^2 + 2x - 3x - 6$
- (A) $(x-3)(x+2)$
- (B) $(x+3)(x-2)$
- (C) $(x+1)(x-6)$
- (D) $(x-1)(x+6)$
- (E) $(x+6)(x-1)$

Q4. Factor the expression: $3x^2 + 2x - 6x - 4$
- (A) $(3x+2)(x-2)$
- (B) $(3x-2)(x+2)$
- (C) $(3x-6)(x+2)$
- (D) $(x-2)(3x+2)$
- (E) $(x+2)(3x-2)$

Q5. Verify the factoring of the expression: $x^2 + 5x + 4x + 20 = (x+4)(x+5)$
- (A) True
- (B) False
- (C) Maybe
- (D) Sometimes
- (E) Not Applicable

Q6. Factor the expression: $2x^2 - 5x - 2x + 5$
- (A) $(2x-5)(x-1)$
- (B) $(2x+5)(x-1)$
- (C) $(x-2)(2x+5)$
- (D) $(2x-1)(x-5)$
- (E) $(2x+1)(x+5)$

Q7. Factor the expression: $3x^2 - 2x + 5x - 10$
- (A) $(3x+5)(x-2)$
- (B) $(3x-2)(x+5)$
- (C) $(x+1)(3x-10)$
- (D) $(3x-5)(x+2)$
- (E) $(x-2)(3x+5)$

Q8. Verify the factoring of the expression: $x^2 - 3x - 2x + 6 = (x-2)(x-3)$
- (A) True
- (B) False
- (C) Maybe
- (D) Sometimes
- (E) Not Applicable

Open Ended Questions (FRQ)

Q9. Factor the expression: $2x^2 + 5x + 3x + 15$

Q10. Verify the factoring of the expression: $x^2 + 4x + 2x + 8 = (x+2)(x+4)$

Chef's Tips
- Double-check calculations
- Watch for sign errors
- Use the distributive property to verify factoring